EX-10 Internetworking with routers in CISCO packet tracer simulator

AIM:

To design and configure a simple internetwork using a router.

PROCEDURE:

Step-1 (Configuring Router 1):

1. Select router and open CLI.
2. Press ENTER to start configuring Router 1.
3. Type enable to activate privileged mode.

Router 1 Command Line Interface

Router > Enable
Router # config t
Enter configuration commands, one per line. End with CNTL/Z. Router (config) # interface FastEthernet 0/0
Router (config-if) # ip address 192.168.10.1 255.255.255.0
Router (config-if) # no shutdown Router (config-if) #
%. LINK-5-CHANGED: Interface GigabitEthernet 0/0, changed state to up Router (config-if) # interface FastEthernet 0/1
Router (config-if) # ip address 192.168.0.1 255.255.255.0
Router (config-if) # no shutdown

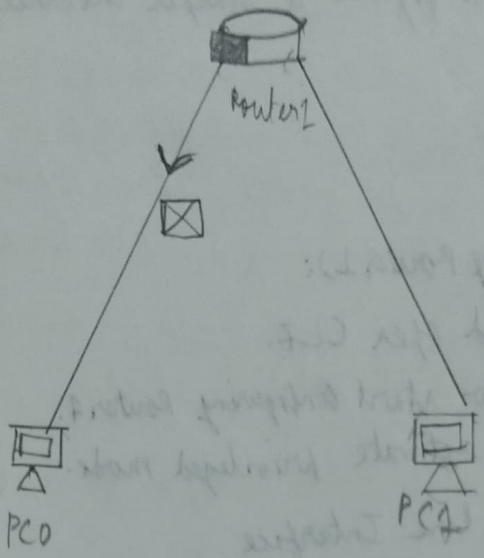
Step-2 (Configuring PCs):

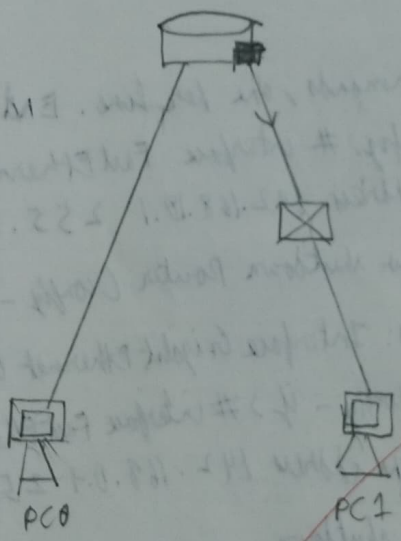
1. Assign IP Addresses to every PC in network.
2. Select PC, Go to Desktop and select configuration and assign an IP address, Default gateway, Subnet mask.
3. Assign default gateway of PC0 as 192.168.10.1
4. Assign default gateway of PC1 as 192.168.20.1

Designed Network Topology

Simulation of Designed Network Topology.

Sending a PDU from PC0 to PC1



Step-3 (Connecting PCs with Routers):

1. Connect Fast Ethernet 0 port of PC0 with Fast Ethernet 0/0 port of Router 1 using a copper straight through cable.
2. Connect Fast Ethernet 0 port of PC1 with Fast Ethernet 0/1 port of Router 1 using a copper straight-through cable.

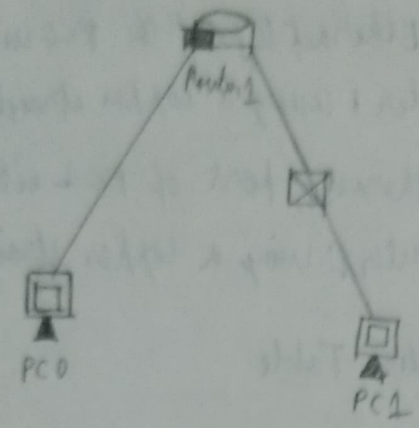
Router Configuration Table:

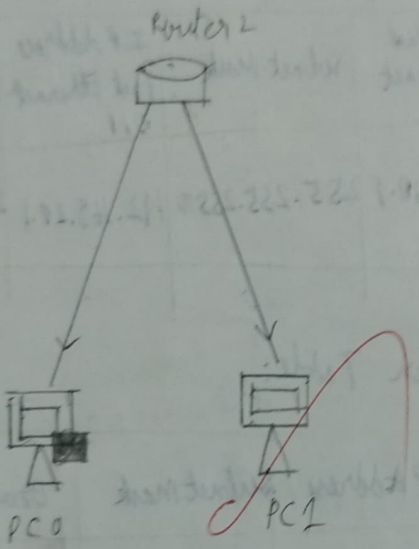
| Device Name | IP Address Fast Ethernet 0/0 | Subnet Mask | IP Address Fast Ethernet 0/1 | Subnet Mask |
|---|---|---|---|---|
| Router 1 | 192.168.10.1 | 255.255.255.0 | 192.168.20.1 | 255.255.255.0 |

PC Configuration Table:

| Device Name | IP Address | Subnet Mask | Gateway |
|---|---|---|---|
| PC0 | 192.168.10.2 | 255.255.255.0 | 192.168.10.1 |
| PC1 | 192.168.10.2 | 255.255.255.0 | 192.168.20.1 |

Acknowledgement from PC1 to PC0



RESULT:

Thus the internetwork using a router is designed and implemented.

Addressing Table

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| PC | Ethernet 0 | DHCP | | 192.168.0.1 |
| Wireless Router | LAN | 192.168.0.1 | 255.255.255.0 | |
| Wireless Router | Internet | DHCP | | |
| cisco.com Server | Ethernet 0 | 209.67.220.220 | 255.255.255.0 | |
| Laptop | Wireless 0 | DHCP | | |

EX 10(b) Internetwork with Wireless Router, DHCP server and Internet Cloud

AIM:

To design and configure an internetwork using wireless router, DHCP server and internet cloud.

PROCEDURE:

Part 1: Build a Simple Network in Logical Topology Workspace

Step 1: Launch Packet Tracer

Step 2: Build the topology

a. Add network devices to the workspace.

Using - device selection box, add network devices to workspace as shown in topology diagram.

b. Change display names of network device

To change display names of network devices click on the device icon on the Packet Tracer Logical workspace, then click on config tab in the device configuration window. Type new name of device in the Display Name box.

c. Add the physical cabling between devices on the workspace.

Part 2: Configure the Network devices

Step 1: Configure the wireless router

a. Create the wireless network on the wireless router.

Click on wireless router icon on the Packet Tracer Logical workspace to open device configuration below In wireless configuration window, click on the GUI tab to view configuration options for wireless router.

Next, click on the wireless tab in GUI to view wireless settings. The only setting that needs to be changed from the defaults is Network Name (SSID). Here type the name "HomeNetwork".

Configure the Internet connection on the wireless router. click on the setup tab in wireless router GUI.

In DHCP Server setting verify that Enabled button is selected and configure static IP address of DNS server at 208.67.220.220.

b: Click on Save settings tab.

Step 2: Configure the Laptop

a. Configure Laptop to access wireless network.

Click on the Laptop icon on the Packet Tracer Logical workspace in Laptop configuration windows select Physical tab.

In the Physical tab you will need to remove Ethernet copper module and replace it with wireless WPC300N module.

With wireless module installed, the next task is to connect laptop to wireless network.

Click on the Desktop tab at top of laptop configuration window and select PC wireless Icon.

Once the Wireless-N Notebook adapter settings are visible, select the Connect tab. The wireless network "HomeNetwork" should be visible in list of wireless networks.

Select the network, and click on the Connect tab found below Site Information Pane.

Step 3: Configure the PC

a. Configure the PC for wired network.

Click on PC icon on the Packet Tracer logical workspace and select Desktop tab and then IP Configuration icon. In IP Configuration window, select DHCP radio button so that PC will use DHCP to receive an IPV4 address from wireless router. Close the IP configuration window.

Click on the Command Prompt icon. Verify that PC has received an IPV4 address by issuing the ipconfig/all command from the Command Prompt. The PC should receive an IPV4 address in the 192.168.0.x range.

Step 4: Configure the Internet cloud

a. Install network modules if necessary

Click on Internet Cloud icon on the Packet Tracer logical workspace and then click on the Physical tab. The cloud device will need two modules if they are not already installed. The PT-CLOUD-NM-1CX which is for the cable modem service connection and PT-CLOUD-NM-1CFE which is for a copper Ethernet cable connection. If these modules are missing, power off the physical cloud device by clicking on the power button and drag each module to an empty module port on device and then power device back on.

b. Identify the From and To ports

Click on the Config tab in cloud device window. In left pane click on Cable under CONNECTIONS. In first drop down box choose coaxial and in second drop down box choose Ethernet then click the Add button

to add there in the From Port and To Port

c. Identify type of provider

While still in Config tab click Ethernet under INTERFACE in left pane. In the Ethernet Configuration window select Cable as the Provider Network.

Step 5: Configure the Cisco.com server

Click on the Cisco.com server icon on Packet Tracer Logical workspace and select services tab. select DHCP from the SERVICES list in the left pane.

In the DHCP configuration window, configure a DHCP with following settings.

- Click ON to turn the DHCP service on

Pool name: DHCPpool
Default Gateway: 208.67.220.220
DNS Server: 208.67.220.220
Starting IP Address: 208.67.220.1
Subnet Mask: 255.255.255.0
Maximum Number of Users: 50

Click Add to add the pool.

Configure the Cisco.com server as a DNS server to provide domain name to IPV4 address resolution

While still in Services tab, select DNS from SERVICES listed in the left pane.

Configure DNS service using following settings

- Click On to turn the DNS service on
- Name: Cisco.com
- Type: A Record
- Address: 208.67.220.220

Click Add to add the DNS service settings

c. Configure the Cisco.com server Global settings
Select the Config tab.
Click on settings in left pane.
Configure the Global settings of the server as follows.
- Select Static
- Gateway: 208.67.220.1
- DNS Server: 208.67.220.220

d. Configure Cisco.com server FastEthernet0 Interface settings
Click on FastEthernet in left pane of Config tab
Configure the Fast Ethernet Interface settings of server as follows:
- Select Static under IP configuration.
- IP Address: 208.67.220.220
- Subnet Mask: 255.255.255.0

Part 3: Verify Connectivity

Step 1: Refresh the IPv4 settings on the PC

a) Verify that PC is receiving IPV4 configuration information from DHCP.

Click on the PC on the Packet Tracer logical workspace and then select Desktop tab of PC configuration window.

Click on the Command Prompt icon.

In the Command Prompt refresh the IP settings by issuing commands ipconfig/release and then ipconfig/renew. The output should show that PC has an IP address in 192.168.0.X range, a subnet mask, a default gateway and DNS server address.

b) Test connectivity to cisco.com server from the PC.
From command prompt, issue command
ping cisco.com. It may take few seconds for the ping
to return. Four replies should be received.

RESULT:

Thus the internetwork using wireless router, DHCP ~~router~~ server and internet cloud. is designed and configured successfully.